



STATISTICS FUNDAMENTALS

From Descriptive Data to Statistical Inference

Populations & Samples • Data Types • Visualization • Central Tendency & Variability
Distributions • Central Limit Theorem • Estimators

Where We Are Going

Two parts, one goal: make sound, data-driven decisions.



PART 1 — Descriptive Statistics

- Populations vs. samples
- Types of data & levels of measurement
- Graphs & tables that fit each data type
- Mean, median, mode & skewness
- Variance, standard deviation, CV
- Covariance & correlation
- **Case study: a real estate database**



PART 2 — Inferential Statistics

- What a distribution really is
- The normal distribution
- Standardization & z-scores
- Sampling distributions
- **The Central Limit Theorem**
- Standard error
- Estimators & point estimates

PART 1

Descriptive Statistics

Organizing, summarizing, and honestly describing the data in front of you.



Know your data: population or sample, type, and measurement level



Choose the right picture: tables, bars, pies, Pareto, histograms, scatter plots



Summarize with numbers: center, asymmetry, spread, and relationships

Population vs. Sample

The first question of every analysis: which one am I holding?



POPULATION (N)

- All items of interest to the study
- **Numbers computed from it are parameters**
- Hard to define, hard to observe, costly



SAMPLE (n)

- A subset drawn from the population
- **Numbers computed from it are statistics**
- Faster and cheaper — what you will almost always use



A thought experiment: “all students.” “All students of a university” includes students on campus, at home, abroad, on exchange, part-time, and even enrolled high schoolers — and the list still misses someone. Populations are hard to pin down; statistical tests are designed for the incomplete data of samples.

What Makes a Sample Good?

Two conditions — and a canteen that failed both.



RANDOM

Every member of the population is chosen strictly by chance — everyone has a shot at being selected.



REPRESENTATIVE

The subset accurately reflects the members of the entire population — not just one convenient corner of it.



Case: 50 students interviewed in the university canteen

- Not random — only students at lunch could be picked; most members never had a chance.
- Not representative — it reflects “students who eat at the canteen,” not all students of the university.
- **The fix: random selection from the full student database — which usually needs institutional access.**

Types of Data

Classify first — the type of data decides which statistics and charts are even possible.



CATEGORICAL

- Describes categories or groups
- Car brands: Mercedes, BMW, Audi
- Yes / No answers (“Do you own a car?”)



NUMERICAL

DISCRETE

- Countable, finite — you can list every value
- Number of children, scores, grades
- Pizza prices in pesos: you can pay ₱199.25, not ₱199.253 — the smallest step is 1 centavo

CONTINUOUS

- Infinite, impossible to count
- Weight, height, area, distance, time
- A drop of sweat changes your true weight — the scale just can’t see it

Levels of Measurement

Qualitative (nominal, ordinal) • Quantitative (interval, ratio)



NOMINAL *Qualitative*

Categories with no order. Car brands; the four seasons.



ORDINAL *Qualitative*

Categories with strict order. Lunch ratings: disgusting → unappetizing → neutral → tasty → delicious.



INTERVAL *Quantitative*

Numbers without a true zero. °C and °F — 0° is an artificial, man-made point, so ratios are meaningless.



RATIO *Quantitative*

Numbers with a true zero. Counts, distance, time — and Kelvin: 0 K is where atoms stop moving.

Why temperature trips people up: 10°C vs. 5°C looks “twice as warm” — but the same days are 50°F vs. 41°F. No true zero, no ratios. In Kelvin, ratios are real.

Knowledge Check 1

Think first, then discuss with a seatmate. Answers are in the speaker notes.



Q1. You survey every 10th customer leaving a mall to study “shoppers in this city.” Is this sample random? Is it representative?



Q2. SAT scores can only take specific listed values. Discrete or continuous? What about your exact height?



Q3. Classify: jersey numbers of a basketball team, and temperature in Kelvin — type and level of measurement for each.

Visualizing Categorical Data

Case: a shop selling German cars — Audi, Mercedes, BMW (illustrative units sold).



Frequency distribution table

Category + frequency. The starting point for every chart.



Bar chart

One bar per category. Fastest way to compare groups.



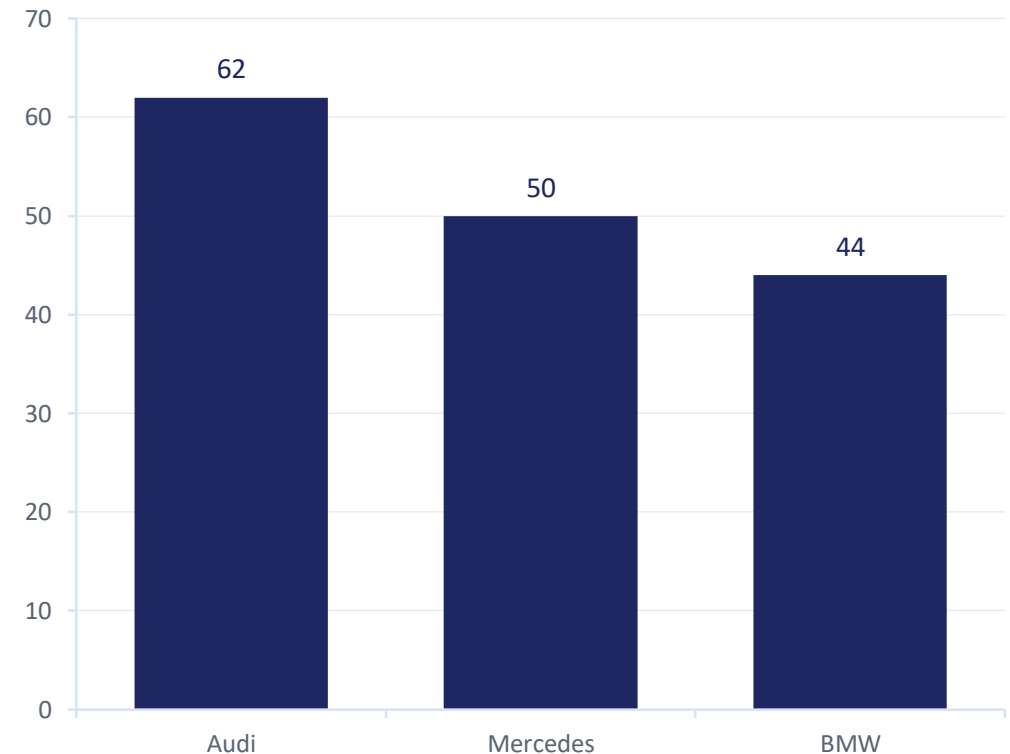
Pie chart

Built on relative frequency (shares sum to 100%). The classic market-share view.



Pareto diagram

Bars in descending order + a cumulative-frequency curve to 100%.



Sorted by frequency — one cumulative line away from a Pareto diagram.

The Pareto Diagram & the 80–20 Rule

A bar chart, sorted — plus one powerful curve.

- **Order categories by descending frequency**
- Add a cumulative frequency curve — the running sum of relative frequencies, ending at 100%
- Read on a second axis (right side); each vertex shows the subtotal of everything to its left
- Combines the strengths of the bar chart (comparison) and the pie chart (share of total)
- Instantly answers: “What do the top 2 categories cover?”



Vilfredo Pareto's principle

80%

of the effects come from 20% of the causes.

Microsoft reported that fixing 20% of software bugs eliminated 80% of the problems customers experienced.

Grouping Numerical Data into Intervals

20 unique numbers = a useless 20-row table. Group first, then plot.

interval width =

largest – smallest
number of intervals

- Aim for 5–20 intervals, depending on data size
- Example: $(100 - 1) \div 5 = 19.8 \rightarrow$ round up to 20 for neatness
- A value joins an interval if it is $>$ lower bound and $\leq$ upper bound
- Relative frequency = interval frequency $\div$ total observations

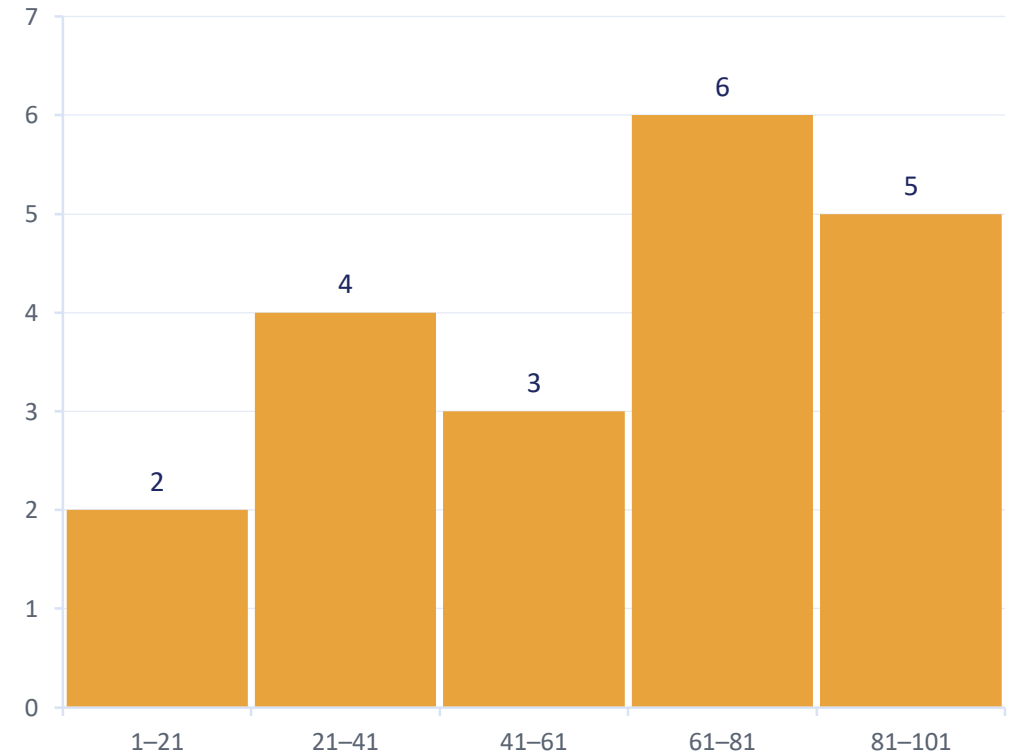
Interval	Frequency	Relative freq.
1 – 21	2	10%
21 – 41	4	20%
41 – 61	3	15%
61 – 81	6	30%
81 – 101	5	25%
Total	20	100%

The worked example from the lecture: 20 numbers, 5 intervals of width 20.

The Histogram

The most common graph for numerical data — a bar chart's cousin, not its twin.

- **Both axes are numerical: intervals on X, frequency on Y**
- Bars touch — each interval ends where the next begins
- Bar charts keep bars apart, because categories are separate things
- Can plot relative frequency (%) instead — same shape, different message
- Unequal intervals exist (survey age brackets like 18–25 ... 60+), but stick to equal widths until you're experienced



Near-zero gaps between bars signal continuity between intervals.

Two Variables at Once

Cross tables for categories, scatter plots for numbers.



Cross (contingency) table

- Two categorical variables: e.g., 3 investors × 3 asset classes (stocks, bonds, real estate)
- **Always add row & column totals — rows give totals per asset class, columns give each investor's holdings**
- Visualize with a side-by-side bar chart to compare within or across investors



Scatter plot

- Two numerical variables: 100 students' SAT reading (X) vs. writing (Y), each point one student
- Read the pattern, not the points: clear uptrend; a dense cluster near the 500-average middle; top students excel at both
- **Jane: 550 reading, 300 writing — an outlier that defies the data set's logic**

Knowledge Check 2

Think first, then discuss with a seatmate. Answers are in the speaker notes.



Q1. Your bar chart of ice-cream flavors has bars touching each other. What's wrong, and why does the convention exist?



Q2. A dataset ranges from 4 to 92 and you want 8 intervals. What interval width would you use?



Q3. In a scatter plot of study hours vs. exam scores, one student studied 40 hours and scored 35%. What do we call this point and how should we treat it for now?

Mean, Median, Mode

One dataset, three answers — NYC pizza prices, with one \$66 posh outlier.



MEAN

\$11

Sum $\div$ count. μ for a population, $\bar{x}$ for a sample. Fragile: one \$66 pizza doubled it.



MEDIAN

\$6

Middle of the ordered list, position $(n+1) \div 2$. With an even n , average the two middle values. Immune to the outlier.



MODE

\$3

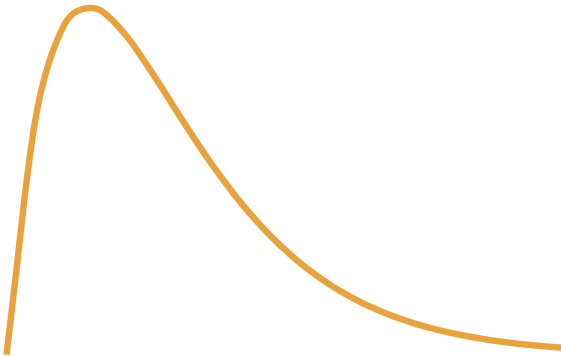
The most frequent value. Works for categories too. No repeats $\rightarrow$ no mode; 2–3 modes are tolerable, more is noise.

There is no single “best” measure — but relying on only one is definitely the worst.

Skewness — Which Way Does the Tail Point?

The direction of the skew follows the TAIL (the outliers), not the bulk of the data.

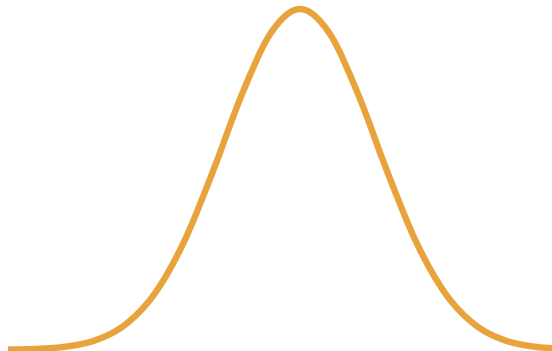
Right (positive) skew



mean > median

Tail & outliers on the right

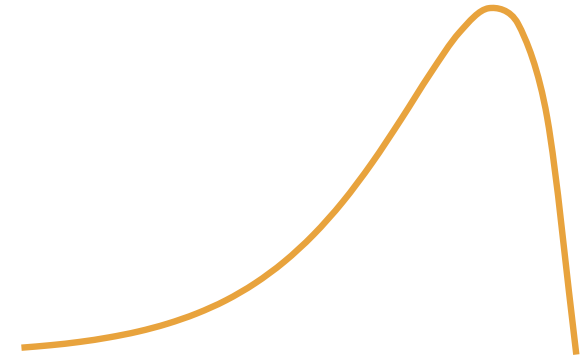
Symmetrical (zero skew)



mean = median = mode

Perfectly centered on the mean

Left (negative) skew



mean < median

Tail & outliers on the left

The mode always marks the highest point. Skewness links central tendency to probability theory.

Variance — Dispersion Around the Mean

Why we square, and why the sample formula divides by $n - 1$.

Population $\sigma^2 = \Sigma(x - \mu)^2 / N$

Sample $s^2 = \Sigma(x - \bar{x})^2 / (n - 1)$

Why square the differences?

- Keeps results non-negative — dispersion is distance, and distance can't be negative. Raw differences would cancel out to nothing.
- Amplifies big deviations: mean 0, observation 100 → squared spread of 10,000.

Worked example: {1, 2, 3, 4, 5}, mean = 3

- **As a population:** $\sigma^2 = 10 \div 5 = 2.0$
- **As a sample:** $s^2 = 10 \div 4 = 2.5$
- The sample version corrects upward — the sample may come from a more spread-out population (e.g., one concentrated at 1s and 5s, with variance 2.96).

Standard Deviation & Coefficient of Variation

Same pizzas, two currencies — and why we need a unitless measure.



Standard deviation = $\sqrt{\text{variance}}$

Back in the original units, so it's directly interpretable — “square dollars” mean nothing, even in statistics. The go-to measure of variability for a single dataset.



CV = standard deviation $\div$ mean

“Relative standard deviation.” Unitless — which makes variability comparable across datasets and across units. Comparing raw standard deviations is meaningless.

Same 11 pizza prices, in dollars vs. Philippine pesos (\$1 = ₱56)

\$3.27

std. dev. in dollars

₱183.12

std. dev. in pesos

0.60

CV in dollars

0.60

CV in pesos

Identical variability, finally visible.

Covariance & Correlation

Direction is easy. Strength needs standardizing.

Covariance — the direction

- > 0 : variables move together (house size & price)
- < 0 : they move in opposite directions
- $= 0$: independent movements
- **Problem: no scale — it can be 0.0023 or 30 million. Unreadable.**

Correlation — the strength

- $r = \text{covariance} \div (\text{std-dev } X \times \text{std-dev } Y)$
- Always between -1 and $+1$ — no exceptions
- House size vs. price: $r = 0.87 \rightarrow$ strong positive
- Ice cream vs. umbrella sales: negative — sun helps one, rain the other



Symmetry. $\text{corr}(X, Y) = \text{corr}(Y, X)$. Size-and-price equals price-and-size — the formula can't tell them apart.



Correlation $\neq$ causation. Size causes price — a house that gets pricier doesn't grow bigger. Regression analysis digs into causality later.

Knowledge Check 3

Think first, then discuss with a seatmate. Answers are in the speaker notes.



Q1. A dataset of salaries has mean ₱45,000 and median ₱32,000. Which way is it skewed, and what likely causes it?



Q2. Sample {2, 4, 6}: compute the sample variance. Why divide by 2 and not 3?



Q3. Dataset A: mean 20, std. dev. 4. Dataset B: mean 400, std. dev. 40. Which is more variable, and which measure told you?

Case Study: A California Real Estate Database

Step 1 — classify every variable before touching a single chart.

Variable	Type	Level	Why
ID	Categorical	Nominal	A number used as a name — the mean ID is meaningless
Age	Numerical	Ratio	True zero at birth; discrete or continuous by task
Age interval	Categorical	Ordinal	Same variable, new level — ordered groups
Price	Numerical	Ratio	Banks treat it as continuous; so do we
Gender	Categorical	Nominal	Binary — like a yes/no question
State	Categorical	Nominal	Labels; numbering states 1–50 changes nothing



Quick test for “numeric-looking” variables: interpret the mean. Mean price = valuable. Mean ID = nonsense → the variable is categorical.

Case Study: What the Numbers Told Marketing

Every descriptive tool from Part 1, working together on one business question.

68%

of US sales come from California; CA + NV cover the top 75% of locations (Pareto)

71%

of buyers were 26–55 years old; peak bracket 36–45 (histogram)

46 ± 13

mean age $\pm$ std. dev.; median 45, mode 48 $\rightarrow$ mean $\approx$ median, slight right skew

-0.17

correlation of age vs. price — below the $|0.2|$ bar, so age doesn't drive spending

- Gender looked male-heavy, but only contract signers are recorded (families buy together) — interesting, not decision-grade. 9% of sales were corporate clients.
- **Existing customers are technically a population; because we're predicting future buyers, we treat the data as a sample and use sample formulas.**

PART 2

Inferential Statistics

Using probability theory and distributions to predict population values from sample data.



Distributions: what they are and the one that rules them all



The normal distribution, standardization, and the Central Limit Theorem

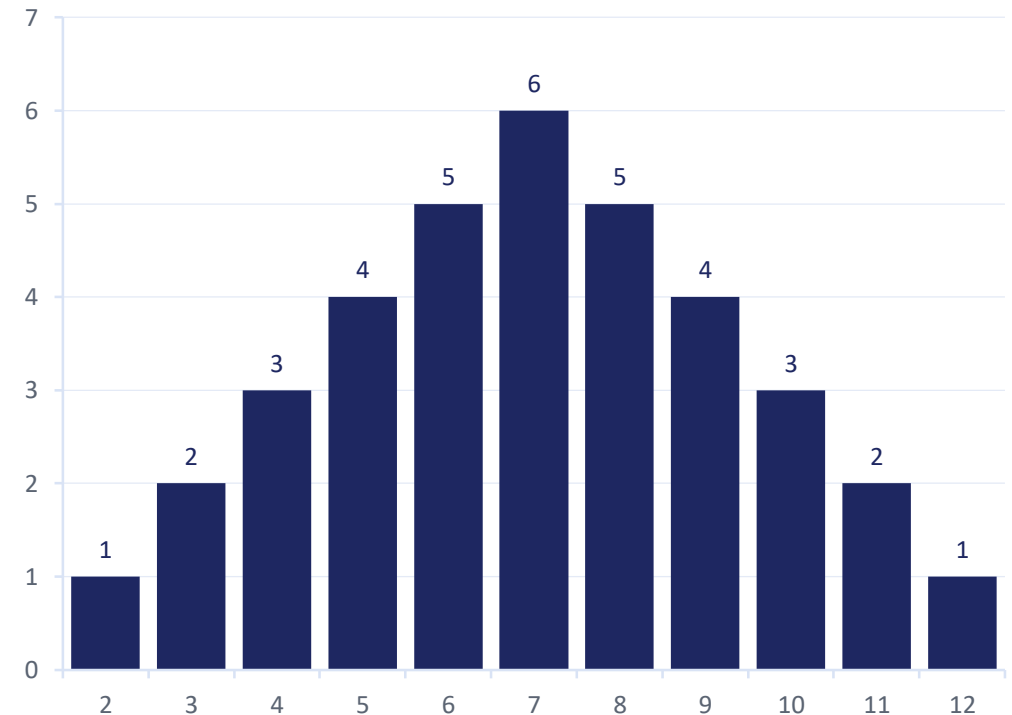


Standard error, estimators, and point estimates — the road to confidence intervals

What Is a Distribution?

A function showing every possible value of a variable — and how often each occurs.

- One fair die: each face has probability $1/6$ (≈ 0.17); a seven has probability 0
- **A distribution covers ALL possible values — complete when probabilities sum to 1 (100%)**
- Equal chances for every outcome = the discrete uniform distribution
- **The graph is only a picture of the distribution — the distribution IS the underlying probabilities**
- Two dice: 36 combinations; guessing the sum is a game of unequal chances →

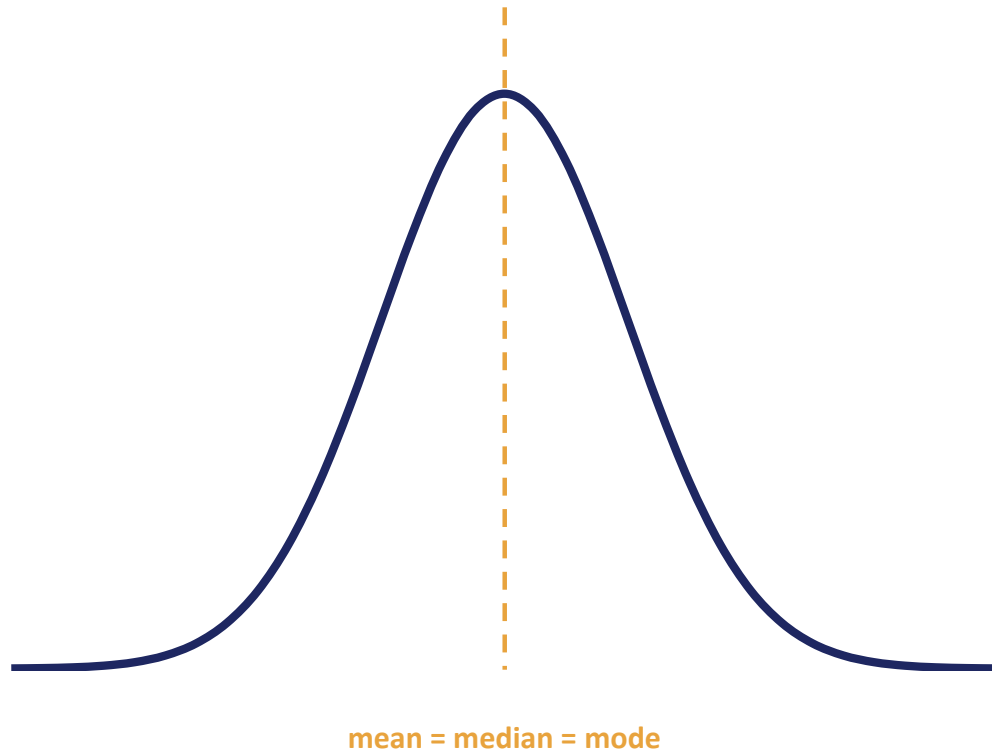


Ways to roll each sum of two dice (out of 36). Seven is king: $6/36 \approx 0.17$.

So far: discrete examples. Inference mostly runs on continuous distributions — starting with the normal.

The Normal (Gaussian) Distribution

The bell curve — the most common distribution in statistics, denoted $N \sim (\mu, \sigma^2)$.



- **Perfectly symmetrical, zero skew, centered on the mean**
- Highest point at the mean — which coincides with the mode
- Change the mean (fixed σ): the curve slides left or right, shape intact
- Change σ (fixed mean): the curve reshapes in place —
 - lower $\sigma \rightarrow$ tall & tight, thinner tails
 - higher $\sigma \rightarrow$ flat & wide, fatter tails
- Why we care: it approximates a huge range of variables, its statistics are elegant, and decisions built on it have a strong track record

Standardization & the Z-Score

Any distribution can be rescaled to mean 0, standard deviation 1.

$$Z = (x - \mu) / \sigma$$

- **Standardizing a normal distribution gives the standard normal, denoted Z**
- Step 1 — subtract μ from every point: the curve slides so its center is 0; shape (and σ) unchanged
- Step 2 — divide every point by σ : the curve reshapes so its spread is exactly 1
- Result: predictions and inference get dramatically easier

Worked example • mean 3, $\sigma = 1.22$

Original

1, 2, 2, 3, 3, 3, 4, 4, 5

– mean (3)

–2, –1, –1, 0, 0, 0, 1, 1, 2

÷ σ (1.22)

–1.63, –0.82, –0.82, 0, 0, 0, 0.82, 0.82, 1.63

New mean = 0, new $\sigma = 1$ — and note: adding or subtracting a constant never changes the standard deviation.

Sampling Distributions

The used-car problem: every sample tells you a slightly different story.

sample mean

\$2,617

sample mean

\$3,201

sample mean

\$2,844

- One sample mean is a single, shaky approximation of the population mean
- So draw many samples and collect their means into a new dataset
- **That dataset's distribution is the sampling distribution of the mean**
- The means cluster around one value — the population mean itself ($\approx \$2,800$ here)
- Averaging the sample means gives a very precise approximation of μ

The Central Limit Theorem

The discovery that revolutionized statistics.

No matter the shape of the population — skewed, uniform, exponential, anything — the sampling distribution of the mean approximates a normal distribution.

μ

the sampling distribution's mean equals the population mean

σ^2/n

its variance is the population variance over the sample size — bigger samples, tighter results

$n \geq 30$

the usual rule of thumb for the CLT to apply

Why it matters: we can run tests, build confidence intervals, and make inferences with the normal distribution — even when the population itself is nothing like normal. The car prices were right-skewed; their sample means were normal anyway.

Standard Error

The standard deviation of the sampling distribution — variability of the sample means.

$$SE = \sigma / \sqrt{n}$$

- Comes straight from the CLT: variance $\sigma^2/n \rightarrow$ take the square root
- **Tells you how well your sample mean approximates the true mean**
- Appears in almost every statistical test — that's why it earned its own name
- **Shrinks as n grows: bigger samples $\rightarrow$ better approximations of the population**

Feel it with numbers ($\sigma = 100$):

$n = 25 \rightarrow SE = 20$

$n = 100 \rightarrow SE = 10$

$n = 400 \rightarrow SE = 5$

Estimators & Point Estimates

“Estimators are like judges — we want them unbiased and efficient.”

The vocabulary

- Estimator: a rule for approximating a population parameter from sample data alone
- Point estimate: a single number ($\bar{x}$ estimates μ ; s^2 estimates σ^2)
- **It sits exactly at the center of a confidence interval — which carries more information**

The two properties

- Unbiased: expected value equals the true parameter
- Efficient: among unbiased estimators, the one with the smallest variance
- “Statistic” is the broader term — every point estimate is a statistic



The “plus one foot” estimator

“Estimate the average American height as the sample mean plus one foot.” It is an estimator — just a biased one, with a bias of exactly one foot. The sensible, unbiased estimator is the sample mean itself.

Knowledge Check 4

Think first, then discuss with a seatmate. Answers are in the speaker notes.



Q1. A population of incomes is heavily right-skewed. You repeatedly draw samples of $n = 50$ and record each mean. What shape does the distribution of those means take, and why?



Q2. $\sigma = 60$ and $n = 36$. Compute the standard error. What happens to it if n rises to 144?



Q3. A classmate proposes estimating average commute time as “sample mean minus 10 minutes, to be safe.” Evaluate this estimator using the two properties.

Key Takeaways

If you remember six things from this unit, make it these.



Classify before you compute — population or sample, data type, measurement level. Everything downstream depends on it.



Variance → standard deviation → CV: from squared units, to real units, to unit-free comparison.



Match the chart to the data: bars/pies/Pareto for categories, histograms and scatter plots for numbers.



The CLT lets sample means behave normally even when the population doesn't — with mean μ and standard error $\sigma/\sqrt{n}$.



Mean, median, and mode are a team. Outliers bend the mean; skew direction follows the tail.



Correlation caps at ± 1 and is symmetric — and it never, by itself, proves causation.